DBMS – DATABASE RELATIONSHIP ANALYSIS USING JOINS

ShopSphere – Online E-Commerce Database Management System

Objectives:

1. Combine Customer, Product, Orders, and Payment tables.
2. Implement INNER JOIN, LEFT JOIN, and RIGHT JOIN.
3. Retrieve complete order details.
4. Display customer purchase history.
5. Generate multi-table reports.

1. Combining Multiple Tables

The ShopSphere database contains different tables that store customer, product, order, and payment information. These tables can be combined using JOIN operations to retrieve related information and analyze customer purchases.

2. INNER JOIN

INNER JOIN returns only the records that have matching values in both tables.

2.1 Customer and Orders

```
SQL> SELECT C.Customer_ID, C.Customer_Name,
  2         O.Order_ID, O.Order_Date
  3  FROM Customer C
  4  INNER JOIN Orders O
  5  ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        101 Arun Kumar                                           1001
15-SEP-26

        102 Priya Sharma                                         1002
16-SEP-26

        103 Rahul Raj                                            1003
20-SEP-26


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        104 Divya Sri                                            1004
18-SEP-26
```

2.2 Orders and Payment

```
SQL> SELECT O.Order_ID,
  2         O.Order_Date,
  3         PY.Payment_Status
  4  FROM Orders O
  5  INNER JOIN Payment PY
  6  ON O.Order_ID = PY.Order_ID;

  ORDER_ID ORDER_DAT PAYMENT_STATUS
---------- --------- --------------------
      1001 15-SEP-26 Successful
      1002 16-SEP-26 Successful
      1003 20-SEP-26 Failed
      1004 18-SEP-26 Successful
```

2.3 Product and Orders

```
SQL> SELECT
  2      P.Product_ID,
  3      P.Product_Name,
  4      P.Price,
  5      OD.Order_ID
  6  FROM Product P
  7  INNER JOIN Order_Details OD
  8      ON P.Product_ID = OD.Product_ID;

PRODUCT_ID PRODUCT_NAME                                       PRICE
---------- -------------------------------------------------- ----------
  ORDER_ID
----------
       101 Laptop                                            45000
      1001

       102 Mouse                                               500
      1002

       103 Keyboard                                           1200
      1003


PRODUCT_ID PRODUCT_NAME                                       PRICE
---------- -------------------------------------------------- ----------
  ORDER_ID
----------
       104 Mobile Phone                                      18000
      1004
```

3. LEFT JOIN

LEFT JOIN returns all records from the left table and matching records from the right table.

3.1 Display All Customers and Their Orders

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.Customer_Name,
  4      O.Order_ID,
  5      O.Order_Date
  6  FROM Customer C
  7  LEFT JOIN Orders O
  8      ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        101 Arun Kumar                                           1001
15-SEP-26

        102 Priya Sharma                                         1002
16-SEP-26

        103 Rahul Raj                                            1003
20-SEP-26


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        104 Divya Sri                                            1004
18-SEP-26

        105 Karthik S
```

4. RIGHT JOIN

RIGHT JOIN returns all records from the right table and matching records from the left table.

4.1 Display All Orders and Customer Details

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.Customer_Name,
  4      O.Order_ID,
  5      O.Order_Date
  6  FROM Customer C
  7  RIGHT JOIN Orders O
  8      ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        101 Arun Kumar                                           1001
15-SEP-26

        102 Priya Sharma                                         1002
16-SEP-26

        103 Rahul Raj                                            1003
20-SEP-26


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT
---------
        104 Divya Sri                                            1004
18-SEP-26
```

5. Complete Order Details

```
SQL> SELECT
  2      C.Customer_ID,
  3      C.Customer_Name,
  4      O.Order_ID,
  5      O.Order_Date,
  6      P.Product_ID,
  7      P.Product_Name,
  8      P.Price
  9  FROM Customer C
 10  INNER JOIN Orders O
 11      ON C.Customer_ID = O.Customer_ID
 12  INNER JOIN Order_Details OD
 13      ON O.Order_ID = OD.Order_ID
 14  INNER JOIN Product P
 15      ON OD.Product_ID = P.Product_ID
 16  ORDER BY O.Order_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_ID PRODUCT_NAME
--------- ---------- --------------------------------------------------
     PRICE
----------
        101 Arun Kumar                                           1001
15-SEP-26        101 Laptop
     45000

        102 Priya Sharma                                         1002
16-SEP-26        102 Mouse
       500

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_ID PRODUCT_NAME
--------- ---------- --------------------------------------------------
     PRICE
```

```
        103 Rahul Raj                                            1003
20-SEP-26        103 Keyboard
      1200

        104 Divya Sri                                            1004
18-SEP-26        104 Mobile Phone

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_ID PRODUCT_NAME
--------- ---------- --------------------------------------------------
     PRICE
----------
     18000
```

6. Customer Purchase History

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         O.Order_ID,
  4         O.Order_Date,
  5         P.Product_Name,
  6         P.Price
  7  FROM Customer C
  8  INNER JOIN Orders O
  9  ON C.Customer_ID = O.Customer_ID
 10  INNER JOIN Order_Details OD
 11  ON O.Order_ID = OD.Order_ID
 12  INNER JOIN Product P
 13  ON OD.Product_ID = P.Product_ID
 14  ORDER BY C.Customer_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_NAME                                          PRICE
--------- -------------------------------------------------- ----------
        101 Arun Kumar                                            1001
15-SEP-26 Laptop                                               45000

        102 Priya Sharma                                          1002
16-SEP-26 Mouse                                                  500

        103 Rahul Raj                                             1003
20-SEP-26 Keyboard                                              1200


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_NAME                                          PRICE
--------- -------------------------------------------------- ----------
        104 Divya Sri                                             1004
18-SEP-26 Mobile Phone                                         18000
```

7. Customer Purchase History with Payment

```
SQL> SELECT C.Customer_Name,
  2         O.Order_ID,
  3         O.Order_Date,
  4         P.Product_Name,
  5         PY.Payment_Status
  6  FROM Customer C
  7  INNER JOIN Orders O
  8  ON C.Customer_ID = O.Customer_ID
  9  INNER JOIN Order_Details OD
 10  ON O.Order_ID = OD.Order_ID
 11  INNER JOIN Product P
 12  ON OD.Product_ID = P.Product_ID
 13  INNER JOIN Payment PY
 14  ON O.Order_ID = PY.Order_ID
 15  ORDER BY C.Customer_Name;

CUSTOMER_NAME                                       ORDER_ID ORDER_DAT
--------------------------------------------------- ---------- ---------
PRODUCT_NAME                                       PAYMENT_STATUS
--------------------------------------------------- --------------------
Arun Kumar                                               1001 15-SEP-26
Laptop                                             Successful

Divya Sri                                                1004 18-SEP-26
Mobile Phone                                       Successful

Priya Sharma                                             1002 16-SEP-26
Mouse                                              Successful


CUSTOMER_NAME                                       ORDER_ID ORDER_DAT
--------------------------------------------------- ---------- ---------
PRODUCT_NAME                                       PAYMENT_STATUS
--------------------------------------------------- --------------------
Rahul Raj                                                1003 20-SEP-26
Keyboard                                           Failed
```

8. Multi-Table Business Report

```
SQL> SELECT C.Customer_ID,
  2         C.Customer_Name,
  3         O.Order_ID,
  4         O.Order_Date,
  5         P.Product_Name,
  6         P.Price,
  7         PY.Payment_Status
  8  FROM Customer C
  9  INNER JOIN Orders O
 10  ON C.Customer_ID = O.Customer_ID
 11  INNER JOIN Order_Details OD
 12  ON O.Order_ID = OD.Order_ID
 13  INNER JOIN Product P
 14  ON OD.Product_ID = P.Product_ID
 15  INNER JOIN Payment PY
 16  ON O.Order_ID = PY.Order_ID
 17  ORDER BY O.Order_ID;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_NAME                                          PRICE
--------- -------------------------------------------------- ----------
PAYMENT_STATUS
--------------------
        101 Arun Kumar                                            1001
15-SEP-26 Laptop                                               45000
Successful

        102 Priya Sharma                                          1002
16-SEP-26 Mouse                                                  500
Successful

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_NAME                                          PRICE
--------- -------------------------------------------------- ----------
PAYMENT_STATUS
--------------------
```

```
        103 Rahul Raj                                             1003
20-SEP-26 Keyboard                                              1200
Failed

        104 Divya Sri                                             1004
18-SEP-26 Mobile Phone                                         18000

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT PRODUCT_NAME                                          PRICE
--------- -------------------------------------------------- ----------
PAYMENT_STATUS
--------------------
Successful
```

9. Payment Transaction Report

```
SQL> SELECT PY.Payment_ID,
  2         PY.Order_ID,
  3         PY.Payment_Status
  4  FROM Payment PY
  5  INNER JOIN Orders O
  6  ON PY.Order_ID = O.Order_ID
  7  ORDER BY PY.Payment_ID;

PAYMENT_ID   ORDER_ID PAYMENT_STATUS
---------- ---------- --------------------
       501       1001 Successful
       502       1002 Successful
       503       1003 Failed
       504       1004 Successful
```

Conclusion:

Thus, the ShopSphere Online E-Commerce Database Management System uses INNER JOIN, LEFT JOIN, and RIGHT JOIN to combine related tables, retrieve complete order details, display customer purchase history, and generate multi-table business and payment reports.